

**VisionInspectAI-**

**Manufacturing Defect Detection & Quality  
Inspection System**

**Milestone-3 Report**

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## **1. Project Overview**

VisionInspect AI is an AI-powered manufacturing quality inspection system developed to automatically detect product defects using anomaly detection techniques. The system integrates Artificial Intelligence with a web-based application to provide real-time inspection results, quality assessment, inspection history, analytics, and reporting.

During Milestone 3, the primary focus was integrating the trained AI model with the backend, connecting it to the frontend, and improving the user interface for inspection, reporting, and history management.

## **2. Work Completed During Milestone 3**

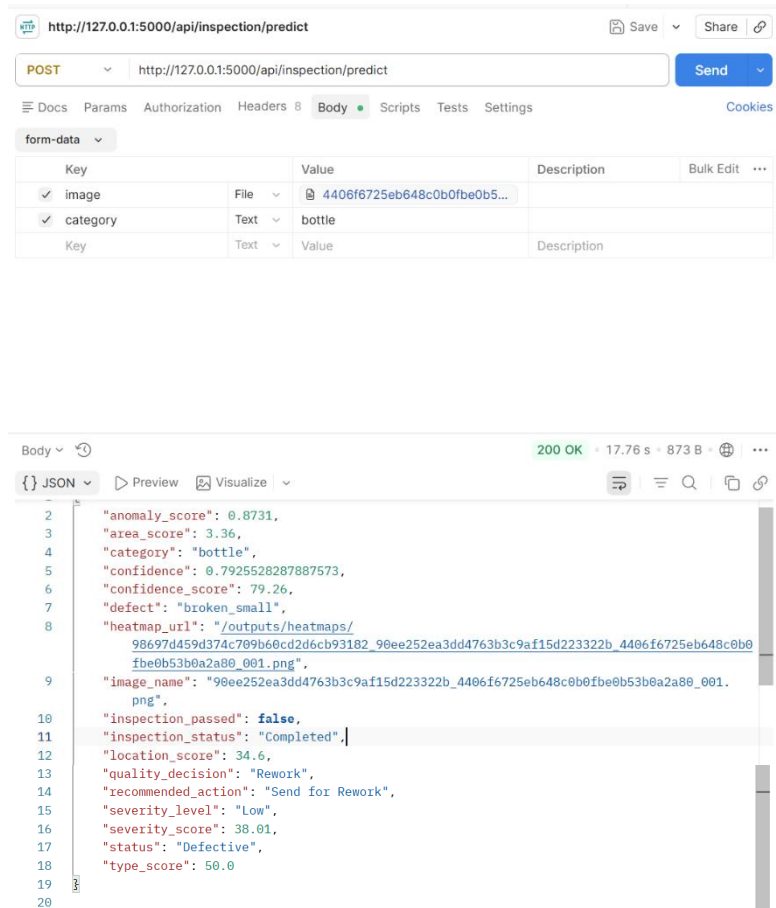
- Integrate the trained AI model with the Flask backend.
- Connect AI inference APIs with the React frontend.
- Display real-time inspection results.
- Generate anomaly heatmaps.
- Re-design the Inspection page UI.
- Connect the Reports module with backend APIs.
- Fine-tune the History page with advanced filtering.
- Improve overall user experience.

## **3. AI Model Integration**

The trained anomaly detection model was successfully integrated with the backend inspection pipeline.

### **Work Completed**

- Connected AI inference with Flask APIs.
- Enabled image upload from the frontend.
- Added image preprocessing before inference.
- Generated anomaly predictions.
- Generated anomaly heatmaps.
- Returned AI prediction results to the frontend.
- Connected quality assessment with inspection results.



**Figure:1 AI Inspection successfully processing upload image. (AI INTEGRATED WITH BACKEND)**

#### 4. Backend API Integration

The existing backend APIs were integrated with the React frontend to enable dynamic data exchange throughout the application.

##### Integrated Modules

- Inspection Prediction
- Analytics Dashboard
- Inspection History
- Reports Dashboard

##### Key Contributions

- Connected frontend components with backend APIs.

- Processed backend responses and displayed them in the user interface.
- Replaced static/demo data with real inspection data.
- Added loading states and error handling for a smoother user experience.

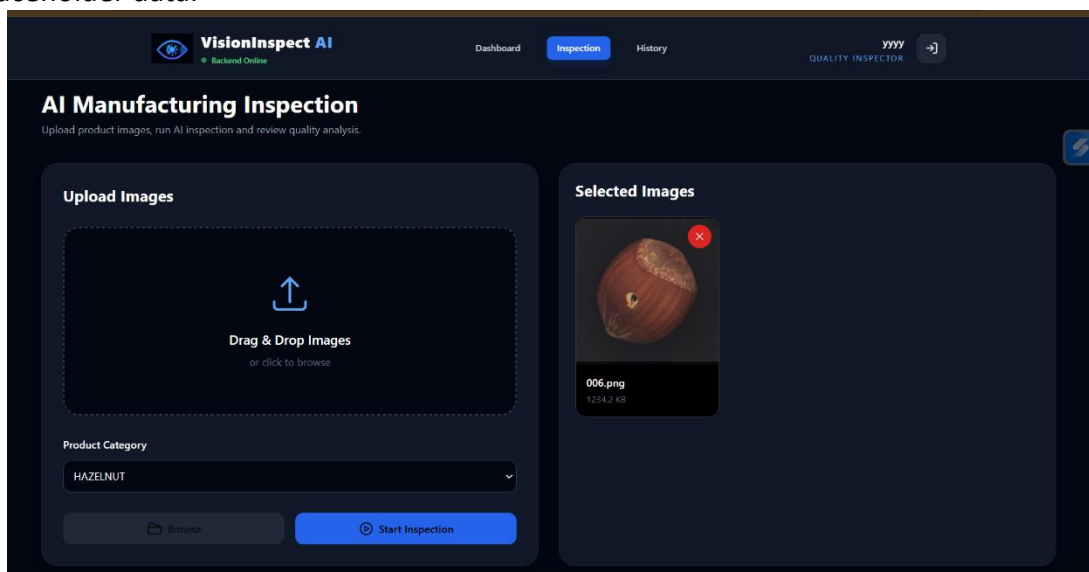
## 5. Inspection Module Enhancement

The Inspection module was redesigned to display real AI-generated inspection results.

### Features Added

- Drag and Drop Image Upload
- Image Preview
- Product Category Selection
- Loading Animation during AI Processing
- Inspection Summary
- Original Image Display
- Heatmap Visualization
- AI Anomaly Score
- Severity Information
- Quality Assessment
- Better UI Components

The inspection page now provides a complete visualization of AI predictions instead of placeholder data.



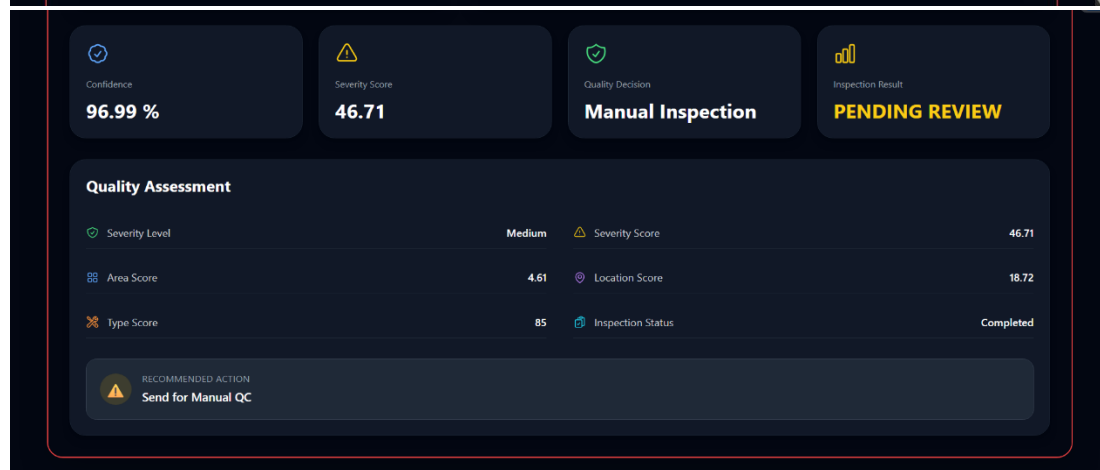
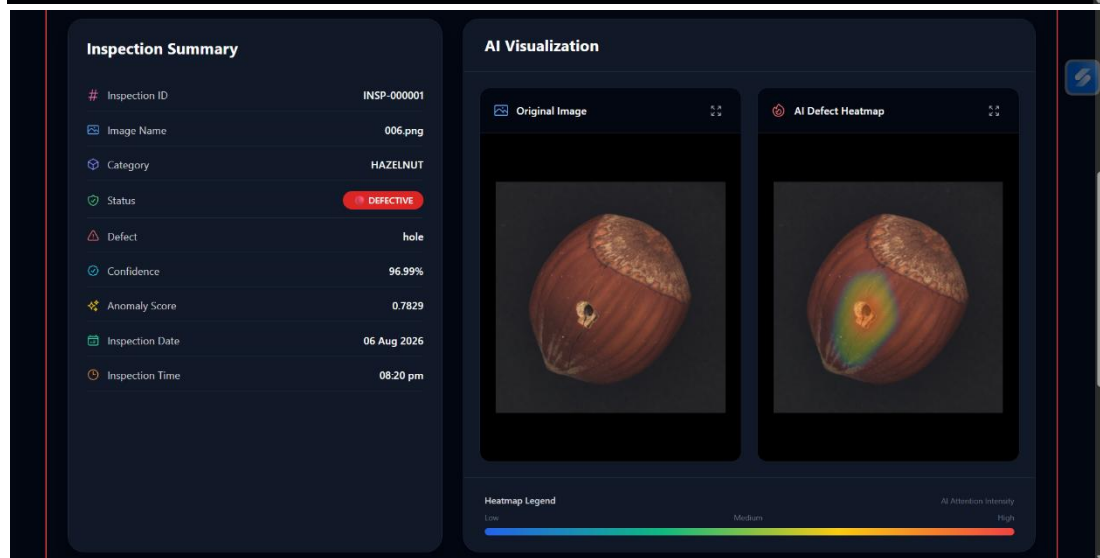
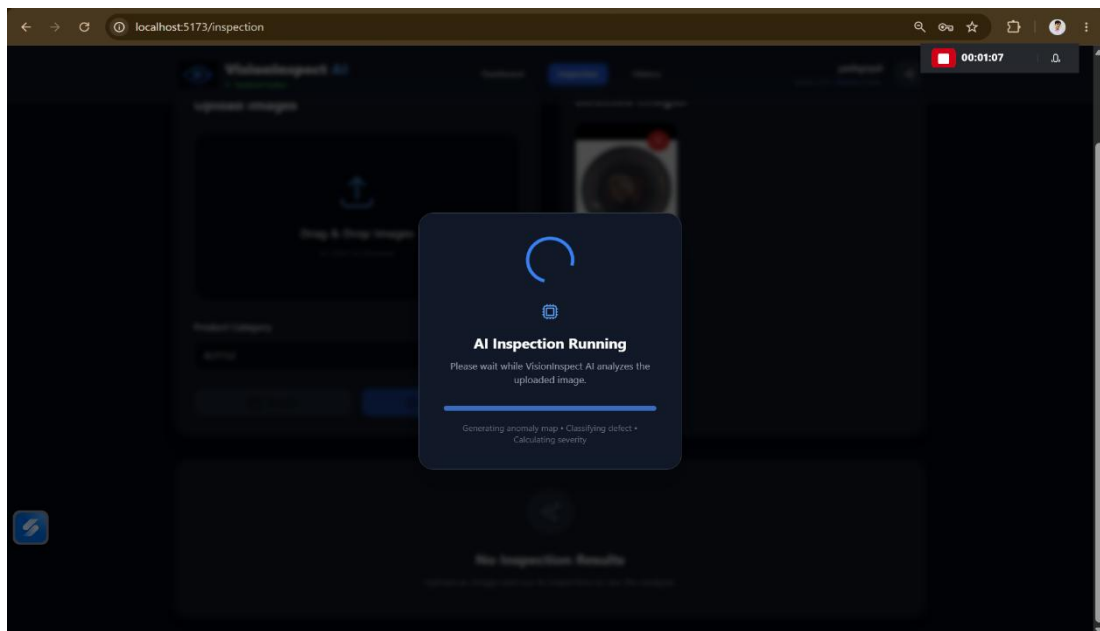


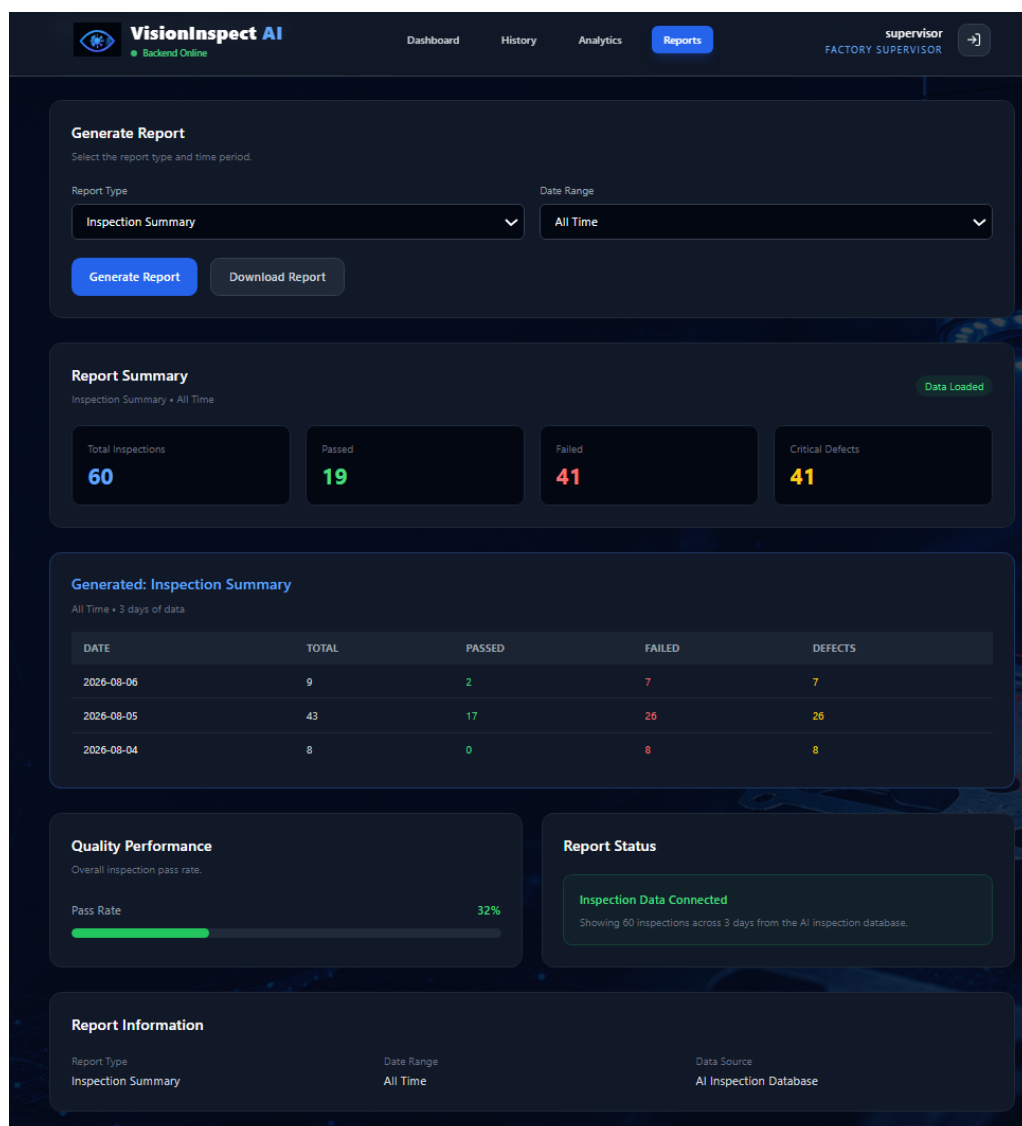
Figure 2: Inspection page displaying AI prediction and generated heatmap.

## 6. Reports Module

The Reports module was connected with backend APIs to display inspection statistics.

### Features

- Inspection Summary
- Report Statistics
- Report Type Selection
- Date Range Selection
- Report Summary Dashboard
- Report Download Support



**Figure 3: Reports Dashboard.**

|    | A  | B         | C       | D        | E        | F         | G        | H         | I       | J         | K        | L        | M        | N         | O      | P           | Q                 | R         | S         | T          | U |
|----|----|-----------|---------|----------|----------|-----------|----------|-----------|---------|-----------|----------|----------|----------|-----------|--------|-------------|-------------------|-----------|-----------|------------|---|
| 1  | Id | inspectio | user_id | username | email    | filename  | category | status    | anomaly | defect    | confiden | severity | severity | quality_d | recomm | heatmap     | inspectio         | inspectio | result_su | created_at |   |
| 2  | 55 | 55        | 6       | yyy      | yash@gm  | 7044a5e6  | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 3  | 54 | 54        | 6       | yyy      | yash@gm  | e5a6a951  | bottle   | Defective | 1       | broken_li | 0.372797 | 34.93    | Low      | Rework    |        | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 4  | 53 | 53        | 6       | yyy      | yash@gm  | f1470679f | bottle   | Defective | 1       | broken_li | 0.372797 | 34.93    | Low      | Rework    |        | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 5  | 52 | 52        | 6       | yyy      | yash@gm  | d876131a  | bottle   | Defective | 1       | broken_li | 0.372797 | 34.93    | Low      | Rework    |        | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 6  | 51 | 51        | 6       | yyy      | yash@gm  | 97fca4f69 | bottle   | Defective | 1       | contamin  | 0.972696 | 51.56    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 7  | 50 | 50        | 6       | yyy      | yash@gm  | f0975d36c | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 8  | 49 | 49        | 6       | yyy      | yash@gm  | 291cc0d5c | carpet   | Normal    | 0.4184  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 9  | 48 | 48        | 6       | yyy      | yash@gm  | 4dd8a796  | leather  | Defective | 1       | cut       | 0.990906 | 41.24    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 10 | 47 | 47        | 6       | yyy      | yash@gm  | 3d8bd659  | leather  | Defective | 1       | cut       | 0.990906 | 61.51    | High     | Reject    |        | Reject Prc  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 11 | 46 | 46        | 6       | yyy      | yash@gm  | 34a01ccf8 | leather  | Defective | 1       | cut       | 0.990906 | 62.42    | High     | Reject    |        | Reject Prc  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 12 | 45 | 45        | 1       | yashgoya | yashgoya | 9253afa0c | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 13 | 44 | 44        | 1       | yashgoya | yashgoya | 93742ebc  | carpet   | Defective | 1       | thread    | 0.586259 | 54.32    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 14 | 43 | 43        | 1       | yashgoya | yashgoya | add92452  | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 15 | 42 | 42        | 3       | yash12   | yashgoya | 3321addt  | bottle   | Defective | 0.7909  | contamin  | 0.965827 | 42.29    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 16 | 41 | 41        | 3       | yash12   | yashgoya | e3101aee  | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 17 | 40 | 40        | 1       | yashgoya | yashgoya | f836426c  | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 18 | 39 | 39        | 1       | yashgoya | yashgoya | 2d186246  | bottle   | Defective | 0.7909  | contamin  | 0.965827 | 42.29    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 19 | 38 | 38        | 1       | yashgoya | yashgoya | 66bfa5b1  | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 20 | 37 | 37        | 1       | yashgoya | yashgoya | d1147ac8  | bottle   | Defective | 0.9536  | contamin  | 0.973599 | 40.88    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 21 | 36 | 36        | 1       | yashgoya | yashgoya | 357f9637c | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 22 | 35 | 35        | 1       | yashgoya | yashgoya | f5c3ff22b | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 23 | 34 | 34        | 1       | yashgoya | yashgoya | dd87b0a   | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 24 | 33 | 33        | 1       | yashgoya | yashgoya | 073a8c1f  | carpet   | Defective | 1       | thread    | 0.586259 | 54.32    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 25 | 32 | 32        | 1       | yashgoya | yashgoya | 90bf68f2t | carpet   | Defective | 1       | broken_li | 0.372797 | 34.93    | Low      | Rework    |        | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 26 | 31 | 31        | 1       | yashgoya | yashgoya | 9ceaf5e   | leather  | Defective | 1       | cut       | 0.990906 | 41.24    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 27 | 30 | 30        | 1       | yashgoya | yashgoya | b218f3bcl | leather  | Normal    | 0.3378  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 28 | 29 | 29        | 1       | yashgoya | yashgoya | 67459a68  | leather  | Defective | 1       | cut       | 0.990906 | 41.24    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 29 | 28 | 28        | 1       | yashgoya | yashgoya | 840e5911  | leather  | Normal    | 0.3378  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 30 | 27 | 27        | 1       | yashgoya | yashgoya | 6cca8750c | leather  | Defective | 1       | cut       | 0.990906 | 41.24    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 31 | 26 | 26        | 1       | yashgoya | yashgoya | 57b02eec  | leather  | Defective | 0.9183  | cut       | 0.93474  | 33.42    | Low      | Rework    |        | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 32 | 25 | 25        | 1       | yashgoya | yashgoya | d5469d32  | leather  | Defective | 1       | cut       | 0.93474  | 60.18    | High     | Reject    |        | Reject Prc  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 33 | 24 | 24        | 1       | yashgoya | yashgoya | 1ccf96952 | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 34 | 23 | 23        | 1       | yashgoya | yashgoya | 2fd28efbf | carpet   | Defective | 1       | thread    | 0.586259 | 54.32    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 35 | 22 | 22        | 1       | yashgoya | yashgoya | 6fe892c9c | carpet   | Normal    | 0.3278  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 36 | 21 | 21        | 1       | yashgoya | yashgoya | e617d5cd  | carpet   | Normal    | 0.3682  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 37 | 20 | 20        | 1       | yashgoya | yashgoya | 46b54cd6  | carpet   | Defective | 1       | hole      | 0.853184 | 45.16    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 38 | 19 | 19        | 1       | yashgoya | yashgoya | d9893a8e  | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 39 | 18 | 18        | 1       | yashgoya | yashgoya | cb84119c  | carpet   | Normal    | 0.2939  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 40 | 17 | 17        | 1       | yashgoya | yashgoya | 75c368db  | carpet   | Defective | 0.9837  | thread    | 0.615781 | 54.91    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 41 | 16 | 16        | 1       | yashgoya | yashgoya | d94181ffe | carpet   | Defective | 1       | cut       | 0.939656 | 61.39    | High     | Reject    |        | Reject Prc  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 42 | 15 | 15        | 1       | yashgoya | yashgoya | c9d85494c | bottle   | Defective | 0.7909  | contamin  | 0.965827 | 42.29    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 43 | 14 | 14        | 1       | yashgoya | yashgoya | f8b5bf2b  | bottle   | Normal    | 0.2262  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 44 | 13 | 13        | 1       | yashgoya | yashgoya | 0e4de571  | bottle   | Normal    | 0.2957  |           |          |          |          | Accept    |        | Ready for / | outputs/Complete  | 1         | Inspectio | #####      |   |
| 45 | 12 | 12        | 1       | yashgoya | yashgoya | be5d21ec  | carpet   | Defective | 0.9837  | thread    | 0.615781 | 54.91    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 46 | 11 | 11        | 1       | yashgoya | yashgoya | 6aa876d0  | cable    | Defective | 1       | poke_ins  | 0.540828 | 53.3     | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 47 | 10 | 10        | 1       | yashgoya | yashgoya | effd8e5f  | cable    | Defective | 1       | cut_outer | 0.515846 | 52.83    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 48 | 9  | 9         | 1       | yashgoya | yashgoya | 144fa2f02 | bottle   | Defective | 1       | broken_li | 0.372797 | 34.93    | Low      | Rework    |        | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 49 | 8  | 8         | 1       | yashgoya | yashgoya | 696554fa  | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Send for I  | /outputs/Complete | 0         | Inspectio | #####      |   |
| 50 | 7  | 7         | 1       | yashgoya | yashgoya | 6e4b9110  | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Inspect A   | /outputs/Complete | 0         | Inspectio | #####      |   |
| 51 | 6  | 6         | 1       | yashgoya | yashgoya | 92c26493c | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Inspect A   | /outputs/Complete | 0         | Inspectio | #####      |   |
| 52 | 5  | 5         | 1       | yashgoya | yashgoya | 72307986c | bottle   | Defective | 0.8609  | contamin  | 0.98601  | 44.02    | Medium   | Manual    | I      | Inspect A   | /outputs/Complete | 0         | Inspectio | #####      |   |
| 53 | 4  | 4         | 1       | yashgoya | yashgoya | c97d790c  | bottle   | Defective | 0.8609  | contamin  | 0.98601  | 44.02    | Medium   | Manual    | I      | Inspect A   | /outputs/Complete | 0         | Inspectio | #####      |   |
| 54 | 3  | 3         | 1       | yashgoya | yashgoya | e4a7ba1c  | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Inspect A   | /outputs/Complete | 0         | Inspectio | #####      |   |
| 55 | 2  | 2         | 1       | yashgoya | yashgoya | 404358c9c | bottle   | Defective | 0.9743  | contamin  | 0.890568 | 40.65    | Medium   | Manual    | I      | Inspect A   | /outputs/Complete | 0         | Inspectio | #####      |   |

Figure 4: Downloaded Report

## 7. History Module Enhancement

The History page was further improved to make inspection records easier to analyze.

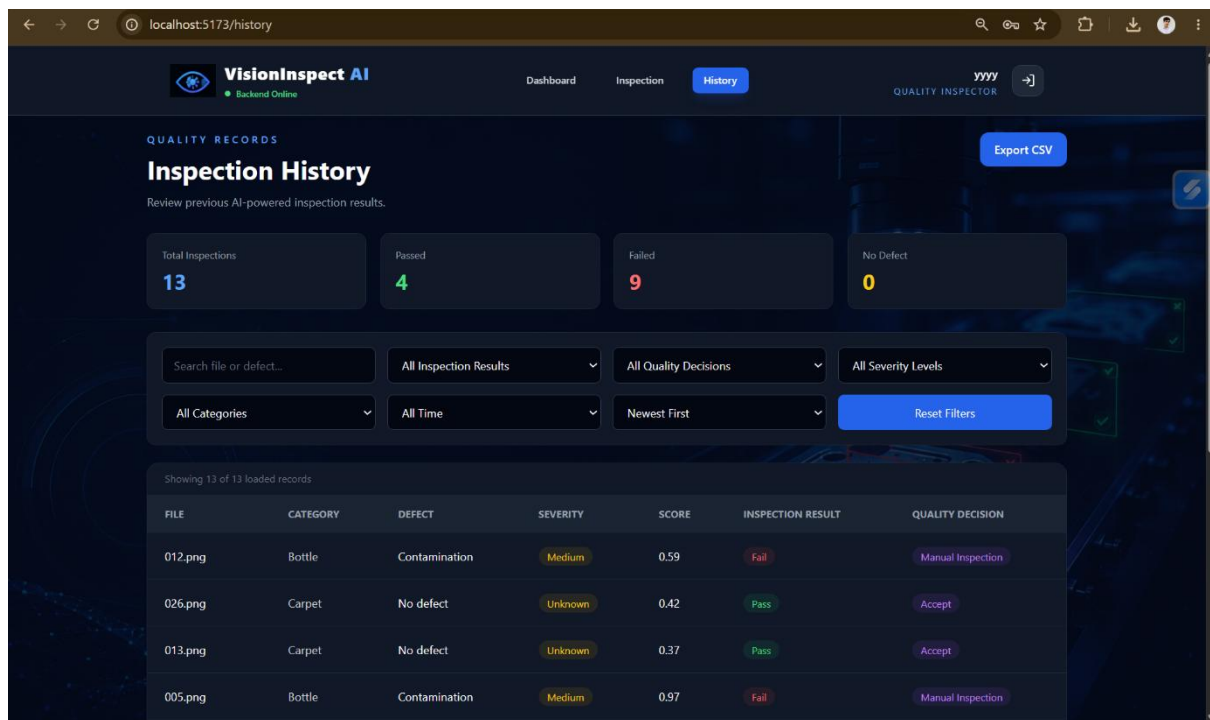
### Enhancements

- Search Functionality
- Category Filter
- Inspection Result Filter
- Quality Decision Filter
- Severity Filter
- Date Range Filter



- Sorting
- Reset Filters
- CSV Export
- Improved Statistics Cards
- Improved Table Layout

These improvements allow users to quickly locate previous inspections and analyze inspection data efficiently.



**Figure 5: Enhanced Inspection History with advanced filtering options.**

## 8. Outcome

At the end of Milestone 3, the VisionInspect AI system successfully supports:

- AI-based defect detection
- Real-time inspection
- Image preprocessing
- Heatmap generation
- Quality assessment
- Analytics
- Inspection history

- Advanced filtering
- Report generation support
- Improved frontend user experience

The application now provides a complete end-to-end inspection workflow from image upload to AI prediction and result visualization.

## **9. Conclusion**

Milestone 3 significantly improved the VisionInspect AI system by integrating the AI model with the backend and frontend, enhancing inspection visualization, strengthening reporting capabilities, and improving inspection history management. The project now offers a robust inspection workflow that enables users to perform quality inspections efficiently while maintaining inspection records and analytical reports.